

Standard Operating Procedure Muffle Furnace

SOP: Fisher Scientific Isotemp Muffle Furnace (Model 650-14)

The Model 650-14 is a high-precision muffle furnace featuring a **PID microprocessor-based controller** and a temperature range of **50°C to 1125°C** (90°F to 2025°F). It utilizes alumina silicate ceramic fiber insulation for rapid heat-up and energy efficiency. This machine is a precision thermal instrument; it is **not explosion-proof** and must never be used with flammable or volatile materials.

Potential Hazards

- **Extreme Thermal Energy**: Internal temperatures reach 1125C. Even after power-off, the hearth plate and chamber walls retain hazardous heat for extended periods.
- **Respiratory Risk**: Alumina silicate ceramic fibers in the chamber can release airborne cristobalite if damaged or during cleaning; this material is a potential carcinogen.
- **Explosion/Fire**: **Strictly Forbidden:** Processing flammable solvents, volatile chemicals, or sealed containers. The unit contains components that can ignite these materials.
- **Electrical**: High-wattage operation (1200 Watts at 120V) requires a dedicated outlet; do not use extension cords.

Personal Protective Equipment (PPE)

PPE Item	Requirement
Eye Protection	Safety Goggles (ANSI Z87.1): Mandatory. A full face shield is highly recommended when opening the door at temperatures above 400°C.
Thermal Protection	Heavy-Duty Heat-Resistant Gloves: Must be rated for temperatures up to 1125°C. Standard welding gloves are insufficient for direct contact with the chamber.
Body Protection	Flame-resistant lab coat and long pants are mandatory to protect against radiant heat and accidental contact with hot surfaces.
Respiratory	N95 Mask: Recommended during cleaning or maintenance. The ceramic fiber insulation can convert to crystalline silica (cristobalite) over 1000°C, which is a known respiratory hazard.



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Before Use

- **Location Check**: Ensure the furnace is on a stable, non-combustible surface and at least **6 inches** away from any walls.
- **Interior Inspection**: Verify all packing materials have been removed. Ensure the **Hearth Plate** is positioned on the floor of the chamber with the **smooth side facing upward**.
- **Venting**: If samples produce odors or non-toxic fumes, ensure the exhaust port is clear or connected to a fume hood via the accessory exhaust tube.
- **Hearth Integrity**: Check for cracks or spills on the hearth plate. A damaged plate can cause uneven heating or damage the bottom heating elements.

During Use

- **Startup**: Place the power switch in the **ON** position. If the "Run" LED is lit, press the RUN key to stop the current program before entering new parameters.
- **Temperature Setup**:
    1. Observe the set temperature in the **Set Display** window.
    2. Use the **UP** and **DOWN** arrow keys to adjust to the desired setpoint.
    3. Press the RUN key to initiate heating. The "Heater" LED will pulse as the PID controller stabilizes.
- **Programming (Ramp/Soak)**: Utilize the MENU key to enter the programming interface for cycles requiring specific heating rates and dwell times.
- **The "200 Rule"**: To prevent thermal shock and safety risks, **do not open the door** until the chamber temperature has dropped below **200°C**.

After Use

- **Critical Cool-Down**: After the cycle is complete, allow the furnace to cool with the door closed until it reaches a safe handling temperature < 50C **Sample Removal**: Always use long-handled tongs and high-temperature gloves. Place hot crucibles on a designated ceramic cooling mat.
- **Cleanup**: Use a brush or vacuum once the unit is at room temperature. **Never use compressed air** for cleaning the chamber as it will aerosolize ceramic fibers.



- **Power Down**: Ensure the power switch is **OFF** and the workstation is clear of all debris.

Emergency Equipment and Procedures

- Emergency Stop: Toggle the main power switch to the OFF position or pull the plug from the outlet if it is safe to reach.
- Controller Failure: If the furnace exceeds the setpoint significantly (Over-Temperature event), disconnect power immediately.
- Reporting: Report all thermal incidents, heating element failures, or refractory damage to the Laboratory Project Supervisor immediately.